

Critical boundary stability with slowly varying density factors

Supplement to research manuscript 008 7 October 2026

Abstract

We extend the critical boundary construction of manuscript 008 to positive slowly varying density factors. For a fixed source with first marginal proportional near zero to $x^{s-1}L(x)e^{-x^2/2}$, $s = p/(p-2)$, we determine the optimal Brenier-map modulus on the unit p th-moment target class for every sufficiently small distance. The answer is an implicit formula determined by $1 + \int_h^a L(x) dx/x$. In particular $L(x) = 1/\log(e/x)$ yields the necessary correction $w^{1/3}[\log \log(1/w)]^{1/(3s)}$. The proof retains the global mass-matching construction of 008 and its explicit all- $\mathcal{P}_2$ potential input from 001 v3.

Scope and provenance. This supplementary research draft to manuscript 008 is not externally peer reviewed or proof-assistant formalized. The multiscale mechanism and overlap interpolation are inherited from [1]; the potential theorem is from [2], which develops the framework of [3]. We prove the additional uniform estimates, normalization and inversion required by the varying factor. No priority claim is made.

1 The source and the exact modulus

Fix $p > 2$, $s = p/(p-2)$, $\beta = s-1 > 0$ and $d \geq 2$. Suppose $L > 0$ is C^2 near zero and, with $\ell = \log L$,

$$x\ell'(x) \longrightarrow 0, \quad x^2\ell''(x) \longrightarrow 0 \quad (x \downarrow 0). \quad (1)$$

Choose a fixed probability density f on $(0, \infty)$ such that, for a sufficiently small $a > 0$,

$$f(x) = cx^\beta L(x)e^{-x^2/2} \quad (0 < x < a), \quad r(x) = f(x_1) \prod_{j=2}^d g(x_j), \quad g(z) = c_g e^{-1/(1-z^2)} \mathbf{1}_{\{|z| < 1\}}. \quad (2)$$

Here $c, c_g > 0$ normalize the densities. The extension of f is chosen so its potential is C^2 and 1-strongly convex on the half-line, with a quadratic tail, as justified below. Write $\rho = r dx$. For its quadratic Brenier maps T_μ , set

$$\mathcal{C}_p = \left\{ \mu : \int |y|^p d\mu(y) \leq 1 \right\}, \quad \Omega(w) = \sup_{\substack{\mu, \nu \in \mathcal{C}_p \\ W_2(\mu, \nu) \leq w}} \|T_\mu - T_\nu\|_{L^2(\rho)}.$$

All constants below may depend on the fixed source, p and d , but not on targets or w . The notation $\asymp$ denotes two-sided bounds by positive constants.

Theorem 1. *For the fixed source (2), put*

$$H(h) = 1 + \int_h^a L(x) \frac{dx}{x}, \quad F(h) = h^{3/2} H(h)^{1/(2s)}, \quad G(h) = h^{1/2} H(h)^{1/(2s)}. \quad (3)$$

The function F is strictly increasing near zero and tends to zero. If $h_w = F^{-1}(w)$ on that interval, then

$$\Omega(w) \asymp G(h_w) = w^{1/3} H(h_w)^{1/(3s)} \quad \text{for every sufficiently small } w > 0. \quad (4)$$

The source has a C^2 potential on its support interior and is strongly log-concave in the extended-valued sense. No full support or regularity across the boundary is asserted. The lower construction has finitely many atoms for each w , with their number allowed to grow as w decreases.

To construct the extension, let $V_0 = x^2/2 - \beta \log x - \ell(x)$. By (1), $V_0'' \geq 1 + \beta/(2x^2)$ on a small interval. Choose $0 < a < a_1$ inside it and a cutoff χ equal to one near a , zero near a_1 , and valued in $[0, 1]$. Continue the curvature as $1 + \chi(V_0'' - 1)$ up to a_1 and as 1 thereafter; integrate twice from $V_0(a)$, $V_0'(a)$. The resulting V is C^2 , has $V'' \geq 1$, and equals $x^2/2 + Ax + B$ eventually. The elementary estimates below give $\ell(x) = o(\log(1/x))$, so $V(x) \rightarrow +\infty$ at zero and e^{-V} is integrable there. Its quadratic tail gives integrability at infinity and all polynomial source moments. Extension by $+\infty$ to $x \leq 0$ makes $V - x^2/2$ proper lower-semicontinuous convex. The transverse potential satisfies $(1/(1 - z^2))'' = 2(1 + 3z^2)/(1 - z^2)^3 \geq 2$ and diverges at its endpoints. Thus ρ is full-dimensional and 1-strongly log-concave. If L is C^∞ , the extension may be chosen C^∞ on the support interior.

2 Slow variation and the upper bound

For every $\eta > 0$, choose $a_\eta > 0$ so that $|x\ell'(x)| \leq \eta$ on $(0, a_\eta)$. Integration gives

$$(x/y)^\eta \leq L(x)/L(y) \leq (y/x)^\eta \quad (0 < x \leq y < a_\eta). \quad (5)$$

It follows that $\ell(x) = o(\log(1/x))$, that $L(ch)/L(h) \rightarrow 1$ uniformly for c in each compact subset of $(0, \infty)$, and, for every fixed $C > 0$,

$$\int_0^{Ch} x^\beta L(x) dx \leq C' h^{\beta+1} L(h). \quad (6)$$

For the last estimate use (5) with $\eta < \beta + 1$ on $(0, h)$ and bounded ratios on (h, Ch) . For any fixed $A > 0$, uniform fixed-ratio convergence gives $H(h) \geq \int_h^{e^{Ah}} L(x) dx/x = (A + o(1))L(h)$. Taking a limsup and then $A \rightarrow \infty$ proves $L(h)/H(h) \rightarrow 0$. Hence

$$\frac{hH'}{H} = -\frac{L}{H} \rightarrow 0, \quad \frac{hF'}{F} \rightarrow \frac{3}{2}, \quad \frac{hG'}{G} \rightarrow \frac{1}{2}. \quad (7)$$

Integration also gives $\log H(h) = o(\log(1/h))$. Thus H is slowly varying, F, G tend to zero and are increasing near zero, and for every fixed $C_0 > 0$,

$$G(F^{-1}(w/C_0)) \asymp G(F^{-1}(w)). \quad (8)$$

Indeed, near zero the logarithmic derivative of F lies between 1 and 2, so its two inverse arguments have a bounded ratio; a bounded logarithmic derivative of G then proves (8). These facts cover finite as well as divergent limits of H .

We state the quantitative inputs precisely. For any full-dimensional $\rho = Z^{-1} e^{-\kappa|x|^2/2 - W(x)} dx$, $\kappa > 0$, with W proper lower-semicontinuous convex, manuscript 001 v3, Theorem 1.1 [2], supplies centered convex potentials $U_\mu \in L^2(\rho)$, finite on the source interior, for all $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, with

$$\|U_\mu - U_\nu\|_2 \leq \sqrt{2/\kappa} W_2(\mu, \nu). \quad (9)$$

Here $\kappa = 1$. The overlap interpolation of [1, Theorem 2.2 and Lemma 2.1], with the density zero-extended, states

$$\|\nabla u - \nabla v\|_2^2 \leq 12dh^{-2} \inf_b \|u - v - b\|_2^2 + 14\Xi_s(h) \sum_j (\|\partial_j u\|_p^2 + \|\partial_j v\|_p^2), \quad (10)$$

$$\Xi_s(h) := \max_{j,\pm} \|(1 - r(\cdot \pm he_j)/r)_+\|_{L^s(\rho)} \leq s \max_j \|r^{1/s}(\cdot + he_j) - r^{1/s}\|_{L^s(dx)}. \quad (11)$$

Losses are set to zero where $r = 0$. The functions u, v are finite convex on an open convex domain supporting the density, their gradients belong to $L^p(\rho)$, and their difference is square-integrable up to a constant. The root inequality follows pointwise from $(1 - t^s)_+ \leq s(1 - t)_+$. The interpolation uses signed-square derivative increments integrated against $\min(r(x), r(x + he_j))$; both translated terms are integrable because their shifted weights are bounded by r . We use these public theorems explicitly, rather than infer (9) from density regularity.

Put $U = f^{1/s}$, extended by zero. On $(0, a)$, $U'/U = (\beta + x\ell' - x^2)/(sx)$, so $|U'|^s \leq CL(x)/x$ since $\beta + 1 = s$. Away from zero the derivative has integrable s th power by the quadratic tail. The contribution to $\int |U(x + h) - U(x)|^s dx$ from $(-h, 2h)$ is at most $C \int_0^{3h} f \leq Ch^s L(h)$. On $x \geq 2h$, the fundamental theorem, Hölder and Tonelli bound it by $h^s \int_{2h}^\infty |U'|^s \leq Ch^s H(h)$. The first term is absorbed because $L/H \rightarrow 0$. Transverse root translations are $O(h)$, as their zero traces and exponential boundary vanishing give $g^{1/s} \in W^{1,s}$. Product integration therefore proves

$$\max_j \|r^{1/s}(\cdot + he_j) - r^{1/s}\|_s \leq ChH(h)^{1/s}.$$

For targets in $\mathcal{C}_p$ the coordinate norm sum in (10) is at most $2d$. Combining (9)–(11) yields

$$\|T_\mu - T_\nu\|_2^2 \leq C\{w^2/h^2 + hH(h)^{1/s}\} \quad (W_2(\mu, \nu) \leq w). \quad (12)$$

Choose the same $h = h_w$ for every pair in the supremum. Since $w^2 = h_w^3 H(h_w)^{1/s}$, this proves the upper half of (4).

3 Uniform mass matching and global convex potentials

We now give the inherited construction with the estimates needed for L . Fix a sufficiently small $a_0 < a/4$ and let

$$t_n = a_0 2^{-n}, \quad R_n = t_n^{-\beta/2}, \quad S_K = \sum_{n=1}^K L(t_n), \quad S_K \asymp H(t_K). \quad (13)$$

The comparison follows by integrating on dyadic intervals, using (5). The fixed upper interval and the added 1 in H are absorbed by $L(t_1) > 0$. It requires neither monotonicity of L nor divergence of S_K .

Let X, Z be independent with densities f, g , and let F_0 be the CDF of X . Set $k = 1/4$ and take $0 < h \leq \epsilon t_K$, with one small fixed $\epsilon > 0$ to be chosen. Define

$$Q_h(y) = \mathbb{E}F_0(y + khZ), \quad l_n = Q_h^{-1}(F_0(t_n - h)), \quad r_n = Q_h^{-1}(F_0(t_n + h)), \quad (14)$$

$$d_n = (l_n + r_n)/2, \quad \gamma_n = (r_n - l_n)/(2h). \quad (15)$$

The inverse is unique at these positive levels. From $F_0(y - kh) \leq Q_h(y) \leq F_0(y + kh)$ one gets $|l_n - (t_n - h)|, |r_n - (t_n + h)| \leq kh$. Uniformly on $|x - t_n| \leq 3h$,

$$f(x) \asymp f(t_n), \quad |f'(x)| \leq Cf(t_n)/t_n,$$

because $f'/f = \beta/x + \ell' - x = O(1/x)$. Symmetry gives $\mathbb{E}Z = 0$, so Taylor's theorem gives $|Q_h(y) - F_0(y)| \leq Ch^2 f(t_n)/t_n$ there, while $Q'_h(y) \geq cf(t_n)$. The inverse mean-value estimate thus proves, uniformly in n, K, h ,

$$l_n = t_n - h + O(h^2/t_n), \quad r_n = t_n + h + O(h^2/t_n). \quad (16)$$

The strips $A_n = \{t_n - h < X < t_n + h\}$ and $C_n = \{l_n - khZ < X < r_n + khZ\}$ have equal mass by the two separate identities

$$\mathbb{E}F_0(l_n - khZ) = F_0(t_n - h), \quad \mathbb{E}F_0(r_n + khZ) = F_0(t_n + h). \quad (17)$$

The first uses symmetry of Z . These equations preserve both neighboring outer masses, as well as the central one. Put $m_n = \rho(A_n) = \rho(C_n)$ and $D_n = \rho(A_n \triangle C_n)$. The coarse bounds imply $|d_n - t_n| \leq kh$ and $1 - k \leq \gamma_n \leq 1 + k$: paired strips overlap and have widths between h and $3h$. All lie within disjoint windows $[t_n - 3h, t_n + 3h]$ if $\epsilon < 1/6$. Comparing endpoints by (16), and replacing density by its value at t_n , gives

$$m_n \asymp hf(t_n), \quad D_n = 2khf(t_n)\mathbb{E}|Z| + O(f(t_n)h^2/t_n).$$

Reduce ϵ once so the error is at most half the positive leading term. Then

$$m_n, D_n \asymp hf(t_n) \quad (18)$$

with constants independent of n, K, h . A fixed sufficiently small bound on h/t_K suffices; that ratio need not tend to zero.

Let $\Phi(x) = \sum_{n=1}^K 2R_n(t_n - x)_+$ and $\tilde{\Phi}(x) = \sum_{n=1}^K 2R_n(d_n - x)_+$. The cusps retain their order. At each cusp let q_n be the midpoint of its adjacent slopes, and $L_n, \tilde{L}_n$ the respective supporting lines. Monotonicity of slopes gives globally $\Phi - L_n \geq R_n|x - t_n|$ and $\tilde{\Phi} - \tilde{L}_n \geq R_n|x - d_n|$, with equality between adjacent cusps. Define finite global convex functions

$$\begin{aligned} U_h(x) &= \max\{\Phi(x_1), L_n(x_1) + R_n h : 1 \leq n \leq K\}, \\ V_h(x) &= \max\{\tilde{\Phi}(x_1), \tilde{L}_n(x_1) + R_n h \gamma_n + R_n h k x_2 : 1 \leq n \leq K\}. \end{aligned} \quad (19)$$

On the source domain, each raised plane dominates exactly on its own strip; the windows are disjoint, so no other plane interferes. Outer gradient atoms are the same ordered slopes in both constructions; central atoms are $q_n e_1$ and $q_n e_1 + R_n h k e_2$. Equations (17), subtracted for adjacent strips, match every outer and central label mass, including the extreme cells. The convex gradients are their Brenier maps: their finite slopes give integrable supporting-plane optimality inequalities.

On a central mismatch the horizontal difference has magnitude R_n ; the vertical difference is $R_n h k$ exactly on C_n . Overlap of paired intervals excludes a direct switch from one outer side to the other. Consequently

$$\begin{aligned} \|\nabla U_h - \nabla V_h\|_2^2 &= \sum_n R_n^2 D_n + k^2 h^2 \sum_n R_n^2 m_n, \\ W_2((\nabla U_h)_\# \rho, (\nabla V_h)_\# \rho)^2 &\leq k^2 h^2 \sum_n R_n^2 m_n. \end{aligned} \quad (20)$$

The second line uses a valid same-label coupling, without asserting its optimality. By (18), both sums have size hS_K , because $R_n^2 f(t_n) \asymp L(t_n)$.

4 Moment normalization and the lower envelope

The jumps are geometric, $R_{n+1}/R_n = 2^{\beta/2} > 1$. Thus every active horizontal slope is bounded by $C \min(x_1^{-\beta/2}, t_K^{-\beta/2})$ on its nonzero region, which lies below a fixed small multiple of a_0 . Shifted cusps change this by only a constant factor, and central vertical slopes obey the same bound since $hk \leq 1$. Using $\beta p/2 = \beta + 1 = s$, the contribution above t_K to the sum of both p th gradient moments is at most $C \int_{t_K}^{C a_0} L(x) dx/x \leq C S_K$. The capped portion below a multiple of t_K is at most $C t_K^{-s} \int_0^{C t_K} x^{s-1} L(x) dx \leq C L(t_K) \leq C S_K$ by (6). Fixed remaining pieces are absorbed by $S_K \geq L(t_1) > 0$.

Scale both potentials by $\lambda_K = (C_* S_K)^{-1/p}$ with one sufficiently large fixed C_* . Each induced target then has p th moment at most one. Write $\Delta_{K,h}$ for the $L^2(\rho)$ distance between their Brenier maps and $w_{K,h}$ for their W_2 distance. Since $1/2 - 1/p = 1/(2s)$, (20) gives

$$\Delta_{K,h} \geq c h^{1/2} S_K^{1/(2s)}, \quad w_{K,h} \leq C h^{3/2} S_K^{1/(2s)}. \quad (21)$$

For every small h , take $K = \lfloor \log_2(a_0 \epsilon/h) \rfloor$. Then $h/\epsilon \leq t_K < 2h/\epsilon$, so the construction is admissible. Equations (13) and the uniform fixed-ratio slow variation of H imply $S_K \asymp H(t_K) \asymp H(h)$. Thus (21) yields $\Delta_h \geq c G(h)$ and $w_h \leq C_0 F(h)$, uniformly for all small h . For any small w , choose $h = F^{-1}(w/C_0)$. This pair belongs to the supremum defining $\Omega(w)$, and (8) proves its lower bound in (4). This constructs a pair for each w ; neither continuity of the integer K nor interpolation between a sequence of distances is needed. The proof of Theorem 1 is complete.

5 A necessary iterated logarithm

Corollary 2. *If $L(x) = [\log(e/x)]^\gamma$ near zero, with any real γ , then*

$$\Omega(w) \asymp \begin{cases} w^{1/3} [\log(1/w)]^{(\gamma+1)/(3s)}, & \gamma > -1, \\ w^{1/3} [\log \log(1/w)]^{1/(3s)}, & \gamma = -1, \\ w^{1/3}, & \gamma < -1 \end{cases} \quad (w \downarrow 0). \quad (22)$$

These are two-sided bounds for every sufficiently small w .

Proof. Writing $t = \log(e/x)$ gives $x \ell' = -\gamma/t$ and $x^2 \ell'' = \gamma(t-1)/t^2$, verifying (1). Direct integration gives $H(h) = 1 + \int_{\log(e/a)}^{\log(e/h)} t^\gamma dt$. Its three regimes are respectively comparable to $[\log(1/h)]^{\gamma+1}$, $\log \log(1/h)$ and 1. Since $\log H(h) = o(\log(1/h))$, $F(h_w) = w$ implies $\log(1/h_w) \sim (2/3) \log(1/w)$. Substitute in (4). $\square$

The case $\gamma = 0$ recovers the critical result of manuscript 008. At $\gamma = -1$ the iterated logarithm is necessary, not merely an upper loss.

Corollary 3 (Exact criterion within the critical family). *For the source (2), the following are equivalent:*

$$\Omega(w) = O(w^{1/3}) \quad (w \downarrow 0), \quad \int_0^a L(x) \frac{dx}{x} < \infty, \quad r^{1/s} \in W^{1,s}(\mathbb{R}^d),$$

where the density root is extended by zero outside its support.

Proof. By (4), $\Omega(w)/w^{1/3} \asymp H(h_w)^{1/(3s)}$, proving the first equivalence. Since $\beta > 0$ and $x \ell' \rightarrow 0$, $|(f^{1/s})'|^s \sim c(\beta/s)^s L(x)/x$ near zero. The derivative is integrable to power s away from zero, and $f^{1/s}$ has zero trace at zero because $\ell(x) = o(\log(1/x))$. Its zero extension therefore lies in $W^{1,s}(\mathbb{R})$ exactly when the displayed integral is finite. Each transverse root already belongs to $W^{1,s}(\mathbb{R})$ with zero trace and L^s norm one. Product integration proves the equivalence with the full root condition. $\square$

This criterion concerns the displayed family, not arbitrary sources. In dimension one, an atomless source has $\|T_\mu - T_\nu\|_2 = W_2(\mu, \nu)$; the obstruction and the dimension restriction here are multidimensional.

Remark 4 (The implicit inverse is essential). In general, $H(h_w)$ cannot be replaced up to constant factors by $H(w^{2/3})$. Take $L(x) = \exp([\log(1/x)]^\alpha)$ near zero, with $1/2 < \alpha < 1$; direct differentiation verifies (1). If $t = \log(1/h)$, substitution and l'Hôpital's rule give

$$H(e^{-t}) \sim \alpha^{-1} t^{1-\alpha} e^{t^\alpha}.$$

Put $t_0 = (2/3)\log(1/w)$ and $t_w = \log(1/h_w)$. Taking logarithms of $F(h_w) = w$ yields $t_w = t_0 + (3s)^{-1} \log H(e^{-t_w})$. Since $t_w \sim t_0$, the preceding asymptotic implies $t_w - t_0 = (3s)^{-1} t_0^\alpha + o(t_0^\alpha)$. Taking logarithms of the asymptotic for H and applying the mean-value theorem therefore gives

$$\log \frac{H(h_w)}{H(w^{2/3})} \sim \frac{\alpha}{3s} t_0^{2\alpha-1} \longrightarrow \infty.$$

Thus that substitution loses two-sided comparability. The logarithmic powers in Corollary 2 are unaffected, as their explicit integration proves.

References

- [1] *Density overlap and a sharp boundary phase diagram for moment-controlled Brenier stability*. mxym/math, research manuscript 008, v1, 7 October 2026. Pinned full manuscript; complete source. Revision [9af06fa4cadaa80e176ed633a7176ed7f2e813db](#).
- [2] *Sharp Brenier stability under target moment bounds*. mxym/math, research manuscript 001, v3, 7 October 2026. Full proof of Theorem 1.1; complete source. Revision [5c6c088aa5abf1c1a4bdca6a8ce5beaa27faaef3](#).
- [3] OpenAI, *Sharp One-Third Stability of Brenier Maps*. OpenAI Math Release preprint, family 374, 25 September 2026. Pinned article. Revision [adc7f1241b42e322a6451854ab7e4b4c146bf78a](#).